

CURRENT ROLE

Marie Skłodowska-Curie Doctoral Fellow at I2SysBIO (CSIC–Universitat de València), in Prof. Ana Conesa's Genomics of Gene Expression lab · Horizon Europe **longTREC Doctoral Network. Research: long-read transcriptomics, isoform regulation, chromatin biology, and **AI for scientific software**.**

EDUCATION

- 2023–

PhD · Bioinformatics — I2SysBIO (CSIC–Universitat de València) · sup. A. Conesa

Valencia, Spain
- 2021–23

MPhil — Cardiff University, School of Engineering · sup. P. Theobald

Cardiff, UK
- 2016–20

BEng · Biology Engineering — Huazhong Agricultural University

Wuhan, China

SELECTED PUBLICATIONS

- 01

2025

T Liu, A Paniagua, F Jetzinger, L Ferrández-Peral, A Frankish, A Conesa. *Transcriptome Universal Single-isoform COnTrol: a framework for evaluating transcriptome reconstruction quality*. **Nature Communications** (in press).
- 02

2025

T Liu, A Conesa. *Profiling the epigenome using long-read sequencing*. **Nature Genetics** 57(1), 27–41.
- 03

2025

C Monzó, T Liu & A Conesa. *Transcriptomics in the era of long-read sequencing*. **Nature Reviews Genetics**.
- 04

2024

FJ Pardo-Palacios, ..., T Liu, et al. *SQANTI3: curation of long-read transcriptomes for accurate isoform identification*. **Nature Methods** 21(5), 793–797.
- 05

2022

T Liu, P Salguero, et al. *PaintOmics 4: tools for integrative analysis of multi-omics datasets*. **Nucleic Acids Research** 50(W1), W551–W559.
- 06

2020

G Ylla, T Liu, A Conesa. *MirCure: a tool for quality control, filtering and curation of microRNAs*. **Bioinformatics** 36(S2), i618–i624.

PROJECTS

- SQANTI-EPI**
LEAD

Protocol-agnostic quality-control framework for long-read epigenomic data — Fiber-seq, m6A and related assays — linking isoform expression to chromatin accessibility.
- AutoRefactor**
FOUNDER

Autonomous AI refactoring agent combining static analysis, automated test generation and mutation testing; safe legacy-code refactoring with Git-native atomic rollback.
- TELLME**
CO-DEVELOP · ELIXIR

Multi-agent LLM service that turns multi-omics networks and enrichment results into citation-grounded narratives (LangChain + GraphRAG + Neo4j).

INVITED TALKS

- 2026

SEBiBC 2026 — Workshop: *AI-assisted improvements in code efficiency and sustainability*

Toledo, Spain
- 2026

ISMB 2026 — *SQANTI-epi*, HiTSeq: High-Throughput Sequencing track (oral)

Washington, DC, USA
- 2025

ISMB/ECCB 2025 — *TUSCO*, iRNA: Integrative RNA Biology track

Liverpool, UK

TEACHING & SERVICE

- 2026

longTREC LURA meeting — participant
- 2026

Long-read Bioinformatics Summer Course — co-organizer & instructor (QC & mapping, SQANTI-reads); *SQANTI-epi* talk

I2SysBIO, València · 25–27 Jun
- 2025

LongTREC Bioinformatics Summer School — instructor & participant

Earlham Inst., Norwich, UK
- 2024

longTREC LURA workshop — co-organizer / coordinator

SKILLS

CODE Python · R · JavaScript · Rust · Bash · SEQ PacBio SMRT · Oxford Nanopore · BIOINFO Nextflow · Bioconductor · Snakemake · Docker · Conda · ML LightGBM · XGBoost · AI LLM apps · multi-agent · prompt eng. · RAG · GraphRAG